



锂硫电池非均相电催化剂

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2022-06-20 收稿, 2022-07-22 修回, 2022-08-24 接受, 2022-08-26 网络版发表

国家自然科学基金(22179089)资助

摘要 锂硫电池具有理论能量密度高、硫资源丰富等特点, 被认为是下一代极具前景的储能电池体系。但锂硫电池中严重的多硫化物“穿梭效应”以及迟缓的反应动力学等问题制约其进一步发展。近年来, 将非均相纳米电催化剂引入锂硫体系, 在抑制“穿梭效应”及提升反应动力学方面取得了显著效果。本文对常见的非均相电催化剂调控策略进行了清晰的分类, 总结了最新的研究进展, 分析了不同调控策略对于促进多硫化物转化的内在机制, 最后对未来锂硫电池电催化剂的发展进行了展望。

关键词 锂硫电池, 非均相电催化剂, 穿梭效应, 反应动力学

1 锂硫电池概述

随着现代工业的快速发展, 对绿色、高效储能系统的需求日益增加^[1-3]。锂硫电池具有较高的理论放电质量比容量(1672 mAh g^{-1})和能量密度(2600 Wh kg^{-1}), 被认为是下一代极具发展前景的先进储能体系之一^[4,5]。如图1(a)所示, 锂硫电池主要由硫正极、隔膜、金属锂负极及有机电解液4部分构成^[6-8]。

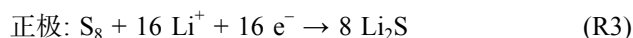
1.1 锂硫电池的电化学原理

与锂离子电池通过锂离子的嵌入脱出实现充放电的工作原理不同, 锂硫电池则依靠正负极发生的氧化还原反应进行充放电^[9-11], 是一个涉及复杂相变和多电子转化的电化学过程^[12]。

如图1(b)所示, 锂硫电池总反应过程可以表示为



以放电过程为例, 正负极反应式如下:



电池放电时, 金属锂负极被氧化生成锂离子, 其通过电解液传输到正极侧与硫分子反应, 同时硫正极得电子生成硫化锂^[13,14]。然而在实际放电过程中, 硫正极发生还原反应会生成一系列多硫化锂(Li_2S_8 、 Li_2S_6 、 Li_2S_4 、 Li_2S_2 、 Li_2S 等), 且经历了固液、液液、液固的多步转化反应^[15]。

具体而言, 反应初期硫发生还原反应, S_8 逐步转化为 Li_2S_4 , 长链的多硫化物转变为短链可溶的多硫化物, 此阶段约贡献了锂硫电池25%的理论容量。在进一步的锂化作用下, 短链可溶的 Li_2S_4 转变为难溶的 Li_2S_2 和 Li_2S , 不导电的 Li_2S_2 和 Li_2S 逐步覆盖电极表面, 产生较大极化电阻使电池电压骤降。此阶段动力学过程相对迟滞, 约贡献了锂硫电池放电总容量的75%^[16]。

1.2 锂硫电池的现存问题

锂硫电池因其具有高的理论能量密度和硫正极材料的低成本及环境友好等特性, 成为当下的研究热点^[17]。但锂硫电池的商业化应用仍面临诸多挑战, 本节从以下3个方面介绍锂硫电池的现存问题和挑战。

引用格式: 朱瑜婕, 史子雄, 孙靖宇. 锂硫电池非均相电催化剂. 科学通报, 2023, 68: 399–413

Zhu Y J, Shi Z X, Sun J Y. Heterogeneous electrocatalysts for lithium-sulfur batteries (in Chinese). Chin Sci Bull, 2023, 68: 399–413, doi: [10.1360/TB-2022-0680](https://doi.org/10.1360/TB-2022-0680)

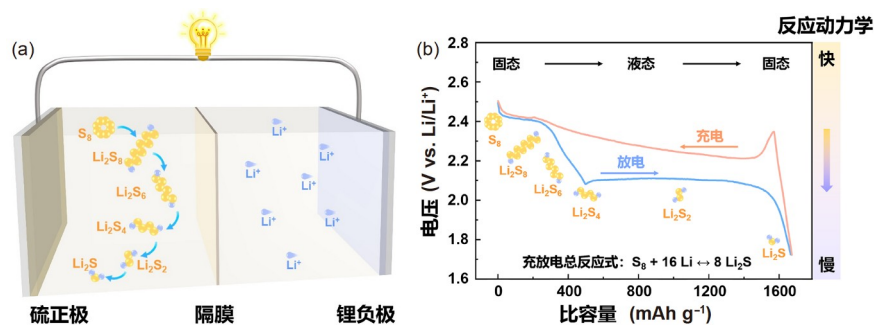


图1 (网络版彩色) 锂硫电池结构示意图(a)和锂硫电池充放电曲线图(b)

Figure 1 (Color online) Schematic of a typical Li-S battery (a) and the charge-discharge voltage profiles of the Li-S battery (b)

(1) 多硫化物的“穿梭效应”。锂硫电池在发生氧化还原反应的过程中,产生的多硫化物在浓度梯度和电场力的作用下,来回穿梭于正负极之间,称之为“穿梭效应”^[18,19]。电池放电时,长链多硫化物从正极侧向负极侧扩散,在负极表面被还原成短链的多硫化锂,进一步形成难溶的Li₂S₂和Li₂S,沉积于锂负极表面形成钝化层,造成硫的利用率下降,电池容量产生不可逆的衰减^[20,21]。

(2) 氧化还原反应动力学缓慢。正极活性物质在充放电过程中,伴随着多种中间体生成,包括一系列可溶的多硫化锂和最终放电产物Li₂S/Li₂S₂。反应初期,硫正极发生还原反应生成可溶的Li₂S₈、Li₂S₆和Li₂S₄中间体。由于该过程反应活化能较低,反应相对容易进行^[14]。随后Li₂S₄还原为不可溶的Li₂S₂/Li₂S,由于Li₂S的本征电子绝缘性,此阶段具有较高的反应势垒,反应速度较慢,因此也被认为是锂硫电池发生电化学反应硫还原过程的决速步骤^[4,22]。

(3) 锂枝晶的生长。电池循环过程中,随着金属锂负极不断沉积和剥离,并且会与电解液发生副反应形成不稳定的固态电解质界面层。由于固态电解质膜的重复破裂生成,电流密度在负极表面分布不均,引起金属锂的不均匀沉积,逐渐形成枝晶。锂枝晶在循环过程中体积无限增大,一方面可能会刺穿隔膜造成电池短路,另一方面锂枝晶也会消耗大量电解液,降低锂金属利用率和库仑效率^[22-24]。值得注意的是,锂硫电池中使用的醚类电解液相较于酯类电解液而言,具有优化锂沉积剥离的效果,并且电解液中的LiNO₃添加剂也可以优化锂负极固态电解质界面层的组分以抑制枝晶生长。

1.3 锂硫电池中电催化剂的设计

在锂硫电池的发展历史中,研究人员为了抑制“穿

梭效应”,曾采用物理限制和化学吸附等方式,实现对于多硫化物的限域和锚定^[25-27]。然而这些方法仍然很难完全解决反应动力学缓慢、循环性能较差等问题。因此,已经广泛应用于多种反应体系的电催化剂被引入到锂硫电池体系,从根本上加速硫反应动力学,抑制“穿梭效应”^[28,29]。电催化剂的引入可以有效降低锂硫反应的活化能,提高电化学反应的速率^[30-32]。

首先,具有极性的电催化剂可以有效吸附多硫化物,抑制其在电解液中的扩散。其次,电催化剂可以有效促进多硫化物的转化,提高活性物质利用率。再次,电催化剂中均匀分布的活性位点还有助于短链多硫化物的均匀沉积。理论体系中电催化剂可分为均相电催化剂和非均相电催化剂。均相电催化剂通常指与可溶性多硫化物在反应体系中相态相同的可溶性有机小分子,而非均相电催化剂则呈现为固体材料。均相电催化剂通常作为氧化还原介体实现催化转化,而非均相电催化剂的催化过程主要包括可溶性多硫化物在催化剂表面发生的吸附、表面反应、脱附等一系列过程^[33]。相比之下,均相电催化剂具有更加明确的催化活性位点,而非均相电催化剂则具有多样的形貌结构以及存在多位点协同催化的可能^[34,35]。本文将从空位调控^[36-38]、掺杂调控^[39-42]、异质结调控^[43,44]、尺寸调控^[45-47]和相调控^[38,48]5个方面,综述近年来非均相电催化剂设计策略和研究进展(图2)。

2 非均相电催化剂的设计策略

2.1 空位调控

空位是指晶格中的原子或离子离开格位后所留下的空格位^[49-51]。通过引入空位缺陷,可以改变电催化剂几何构型和电子结构,而提升电催化剂的催化活

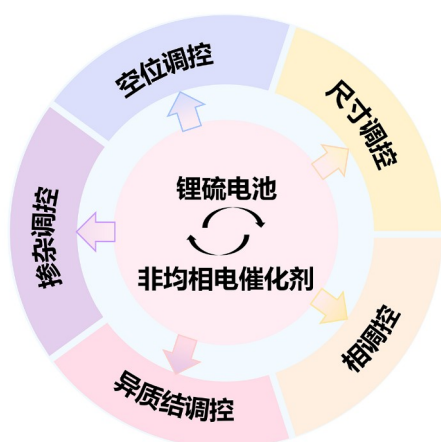


图2 (网络版彩色) 锂硫电池非均相电催化剂设计策略示意图
Figure 2 (Color online) An overview of designing strategies of heterogeneous electrocatalysts in Li-S batteries

性^[52~54]。此外,空位的引入还会降低表面原子的配位数,增大与其他物质的结合能^[55,56]。根据空位种类,本节将从阴离子空位和阳离子空位两个方面,系统介绍通过空位调控设计电催化剂应用于锂硫电池的最新研究进展^[57,58]。

近年来,以氧空位^[53,59,60]、硫空位^[49]、碳空位为代表的阴离子缺陷因其制备工艺简便且催化活性高而受到广泛关注。孙正明课题组^[57]设计了一种用富含氧空位的 Ti_nO_{2n-1} 量子点(oxygen-vacancy-rich Ti_nO_{2n-1} quantum dots, OV- T_n QDs)修饰的多孔碳纳米片(porous carbon nanosheets, PCN)作为锂硫电池的电催化剂,可以有效加强多硫化物的吸附与转化。他们通过对X射线光电子能谱(X-ray photoelectron spectroscopy, XPS)以及电子自旋共振波谱(electron paramagnetic resonance, EPR)的分析,证实了 T_n QDs中氧空位的存在。如图3(a)所示, O 1s的XPS谱图中除了Ti-O键、Ti-OH键和吸附水分子对应的信号外,在532.1 eV处出现了一个特征峰,证明氧空位的存在。EPR测试(图3(b))显示, T_n QDs的 g 值约为2.001。该值进一步证实了 T_n QDs中氧空位的存在,与XPS结果一致。为进一步证实OV- T_n QDs@PCN对多硫化物的吸附作用,他们以 Li_2S_4 为例进行电化学动力学测试和密度泛函理论(density function theory, DFT)计算(图3(c))。结果表明, Li_2S_4 /OV- T_n QDs与 Li_2S_4 /Ti₃C₂T_x相比,具有更高的结合能。由此表明, OV- T_n QDs可有效增强催化剂与 Li_2S_4 等多硫化物之间的相互作用。该工作对引入的氧空位在多硫化物催化转化过程中的作用进行了阐述,并为设计其他富含氧空位

量子点的MXene材料提供了借鉴。

黄佳琦课题组^[58]通过原位蚀刻立方氮化镍获得富含铁空位的电催化剂(cubic nickel-iron nitride, Ni_3FeN),并揭示了缺陷催化剂的原位转化过程。他们以催化活性较差的六方氮化镍(hexagonal nickel nitride, Ni_3N)作为预催化剂,引入阳离子铁掺杂后转化为催化活性较高的 Ni_3FeN (图3(d))。在锂硫电池的反应过程中,由于多硫化物的作用,拐角位置的Fe被刻蚀,因此Ni的周围则留下大量的空位缺陷,使之具有更强的催化活性(图3(e))。他们使用高角度环形暗场扫描透射电子显微镜(high-angle annular dark-field scanning transmission electron microscopy, HAADF-STEM)图像和相应的元素映射对循环前后 Ni_3FeN 颗粒进行分析,显示循环前Ni和Fe在空间分布上偏差极小(图3(f))。然而,在0.5 C的电流密度下循环5圈后的 Ni_3FeN 颗粒中,Fe在边缘处富集,即原始 Ni_3FeN 颗粒在循环过程中与多硫化物发生反应并产生相迁移(图3(g))。这种巧妙设计的催化剂使锂硫电池在大倍率和高负载的条件下,仍具有优异的倍率性能。值得一提的是,在3 C下,比容量可达822 mAh g⁻¹,而 Ni_3N 在相同条件下容量不及其二分之一,表明循环过程中原位形成空位缺陷对与多硫转化反应高效的电催化作用(图3(h))。该工作一方面阐明了阳离子空位在锂硫化学中的重要意义,另一方面也提出了一种原位蚀刻的电催化剂设计策略,为锂硫电池空位电催化剂的设计开辟了新思路。

2.2 掺杂调控

杂原子的引入可以有效改善材料的极性和电子排布,进而提高催化剂的本征活性^[61~63]。本节从碳基材料和金属化合物中的杂原子掺杂两方面具体阐述。

碳材料因其良好的导电性及较大的比表面积被广泛用于锂硫电池中作为正极宿主^[64~66]。为增强碳材料与多硫化物的相互作用,通常通过引入氮^[65,67,68]、硫^[69]、磷(P)^[70]以及贵金属^[71]等杂原子,使之具有更强的极性和对于多硫化物的锚定能力^[72]。Park课题组^[70]基于碳纳米管自支撑导电纤维设计的柔性正极,并将P掺杂的微孔和介孔碳复合物作为修饰隔膜的中层,以实现柔性长续航的高能量密度锂硫电池。他们将相同质量的炭黑(carbon black, SPB)、活性炭(activated carbon, YP)以及P掺杂后的材料(SPBP@P和YP@P)加入到等量的 Li_2S_8 溶液中可视化观察,并进行了紫外可见吸收光谱(ultraviolet-visible, UV-vis)测试(图4(a))。与未

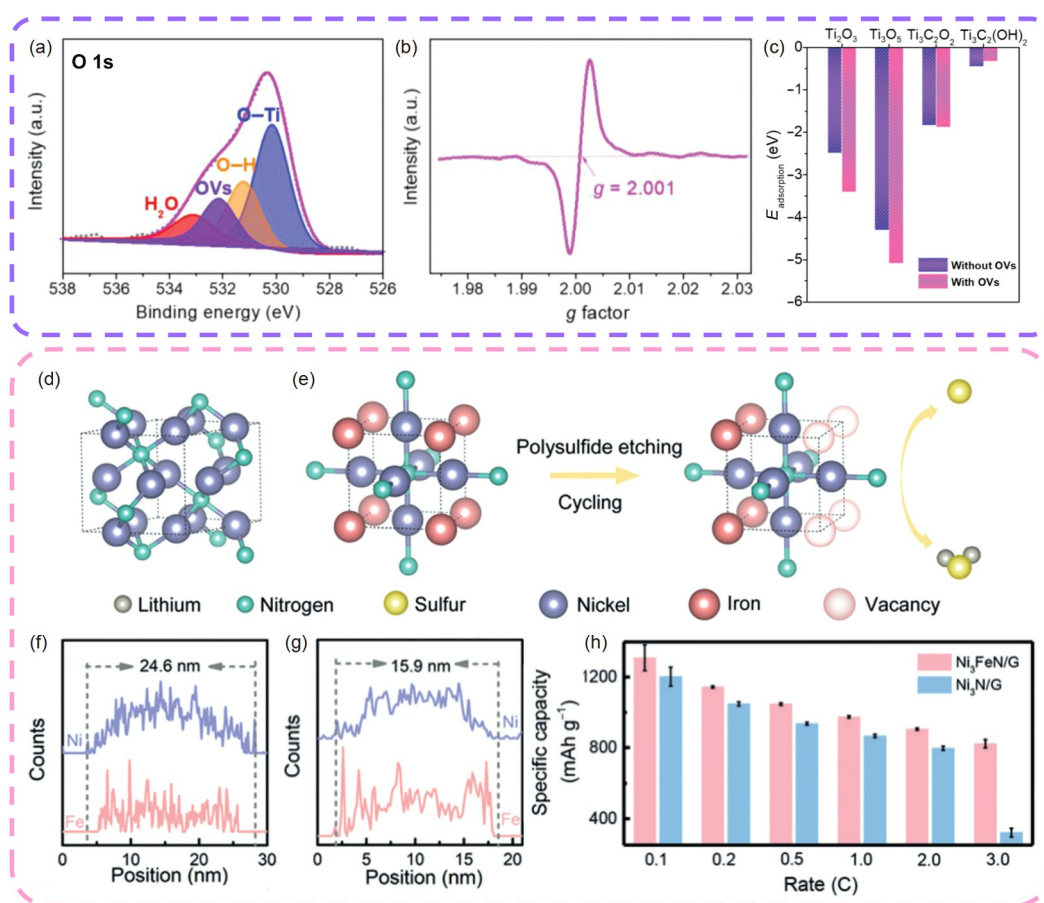


图3 (网络版彩色) 锂硫电池中的空位调控。(a) OV-T_nQDs@PCN的XPS O 1s谱图。(b) OV-T_nQDs@PCN的EPR谱图。(c) Li₂S₄在含氧空位以及不含氧空位的Ti₂O₃ (012), Ti₃O₅ (100), Ti₃C₂O₂ (001)和Ti₃C₂(OH)₂ (001)表面上的吸附能^[57], Copyright © 2021, John Wiley and Sons。(d) 六方Ni₃N的原子结构模型。(e) 立方Ni₃FeN的原子结构模型以及多硫化物刻蚀至活性Ni₃Fe_{1-δ}N相的过程示意图。(f) Ni₃FeN-G的线性元素分布。(g) Ni₃FeN/G在0.5 C下循环5次后的线性元素分布。(h) Ni₃FeN/G以及Ni₃N/G的倍率性能对比^[58], Copyright © 2019, John Wiley and Sons

Figure 3 (Color online) Vacancy regulation in Li-S batteries. (a) O 1s XPS profiles of OV-T_nQDs@PCN. (b) EPR spectra of OV-T_nQDs@PCN. (c) Adsorption energies of Li₂S₄ on Ti₂O₃ (012), Ti₃O₅ (100), Ti₃C₂O₂ (001) and Ti₃C₂(OH)₂ (001) planes with and without oxygen vacancies^[57], Copyright © 2021, John Wiley and Sons. (d) Atomic structure model of hexagonal Ni₃N. (e) Atomic structure model of cubic Ni₃FeN and illustration of the polysulfide etching process toward an active Ni₃Fe_{1-δ}N phase. (f) Linear elemental distribution of Ni₃FeN-G. (g) Linear elemental distribution of Ni₃FeN/G after 5 cycles at 0.5 C. (h) Rate performance of Ni₃FeN/G and Ni₃N/G^[58], Copyright © 2019, John Wiley and Sons

掺杂的材料相比, 加入SPB@P和YP@P的Li₂S₈溶液, 在放置24 h后, 颜色近乎透明。UV-vis光谱显示, SPB@P与YP@P没有出现多硫化物的特征峰(Li₂S₄在420 nm, Li₂S₆在470 nm), 由此可见, 多硫化物几乎完全被捕获。接触角的测试也显示, 修饰后隔膜的润湿性能显著提高, 从而使之具有更优异的电化学活性和更高的离子导电性(图4(b), (c))。此外, 为了验证P掺杂的隔膜对“穿梭效应”的抑制作用, 他们在多硫化物溶液(右侧)和纯溶剂混合物(左侧)之间放置不同的隔膜, 观察其在不同时间下的颜色变化。如图4(d)所示, 静置12 h后, YP@SPB@P修饰的隔膜由于对多硫化锂“穿梭效应”的有效抑制, 左侧溶液没有明显的颜色变化。究其原因, 一方面是由

于YP与SPB的微孔和中孔的物理限制效应, 另一方面是归功于P掺杂与多硫化锂之间的较强相互作用, 实现了对于多硫化锂强化学的锚定。该工作以实现柔性且便携锂硫电池为出发点, 用P掺杂的微孔和介孔碳复合物对隔膜进行修饰, 有效加快反应动力学并提升电池的电化学性能。此外, YP@SPB@P组装的高柔性软包电池在无人机中的应用表明了锂硫电池商用化技术的突破。

作为另一种常见的掺杂方式, 在金属化合物中掺杂金属原子, 也可以有效提升金属化合物的催化活性, 提高对于多硫化物的转化效率, 进而促进硫氧化还原动力学并缓解“穿梭效应”。李喜飞研究组^[73]为了改善

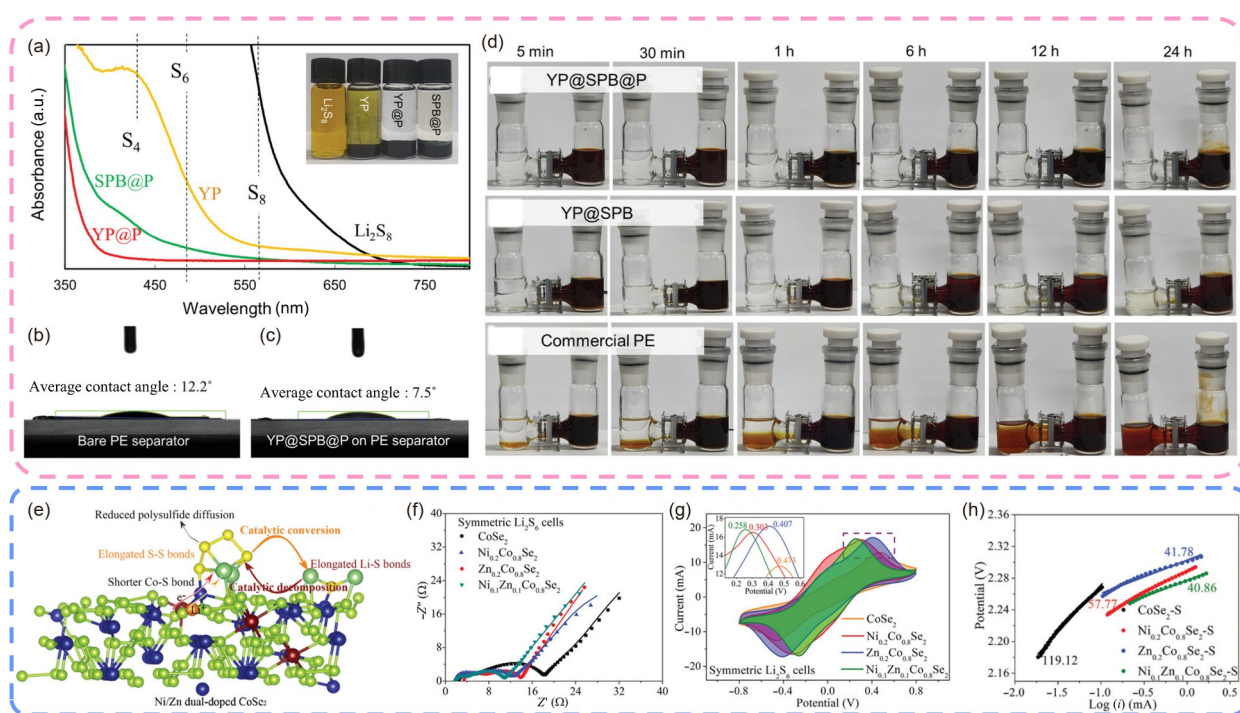


图 4 (网络版彩色) 锂硫电池中的掺杂调控。(a) 不同碳材料对于多硫化物溶液的UV-vis测试曲线。插图为在室温下 Li_2S_8 溶液中加入不同种材料保持24 h后的数码图像。液态电解液对于PE隔膜(b)和YP@SPB@P修饰的PE隔膜(c)的接触角测试。(d) 不同隔膜对多硫化物溶液扩散的阻隔实验^[70], Copyright © 2022, John Wiley and Sons。(e) Ni/Zn双掺的 CoSe_2 催化锂硫电池氧化还原反应示意图。 Li_2S_6 对称电池的电化学阻抗图(electrochemical impedance spectroscopy, EIS)(f)和伏安测试曲线(g)。(h) 不同 CoSe_2 -S电极在氧化过程中的Tafel曲线^[73], Copyright © 2022, John Wiley and Sons

Figure 4 (Color online) Doping regulation in Li-S batteries. (a) UV-vis absorption profiles of the Li_2S_x solutions with different types of carbon materials. Inset: Digital camera image of the Li_2S_8 solutions before and after being maintained at room temperature for 24 h. Contact angle measurement of the liquid electrolyte on bare PE (b) and YP@SPB@P-coated PE (c) separators. (d) Photographs showing the diffusion properties of the Li_2S_x solution through the various separators^[70], Copyright © 2022, John Wiley and Sons. (e) Schematic illustration of Ni/Zn dual-doped CoSe_2 in catalyzing the redox of Li-S batteries. The EIS curves (f) and cyclic voltammetry curves (g) of the symmetric Li_2S_6 cells. (h) The Tafel plots of oxidation processes for different CoSe_2 -S electrodes^[73], Copyright © 2022, John Wiley and Sons

CoSe_2 的催化效应, 设计了一种双功能的镍/锌(Ni/Zn)双掺杂的 CoSe_2 , 能够有效催化多硫化物和 Li_2S 之间的转化反应以及 Li_2S 的分解(图4(e)). 如图4(f)的电化学阻抗图(electrochemical impedance spectroscopy, EIS)所示, 双掺杂的 $\text{Ni}_{0.1}\text{Zn}_{0.1}\text{Co}_{0.8}\text{Se}_2$ 电极具有最小的电荷转移电阻。因此, 电化学反应中多硫化物在 $\text{Ni}_{0.1}\text{Zn}_{0.1}\text{Co}_{0.8}\text{Se}_2$ 界面上显示出最佳的界面反应动力学。为进一步验证双掺杂催化剂的催化作用, 他们以含 Li_2S_6 的电解液组装对称电池, 进行循环伏安(cyclic voltammetry, CV)测试(图4(g)). 所得的曲线中, $\text{Ni}_{0.1}\text{Zn}_{0.1}\text{Co}_{0.8}\text{Se}_2$ 电极的电流响应最大。这表明双掺杂 CoSe_2 在促进多硫化锂转化方面表现出更优异的催化活性。其中, 与Zn掺杂相比, Ni掺杂的 CoSe_2 在改善多硫化锂转化动力学方面起着主要作用。除了放电阶段多硫化物的转化之外, 充电过程中 Li_2S 的分解转化也是影响锂硫电池氧化还原反应动

力学的关键因素。负载硫之后, 不同 CoSe_2 -S电极在氧化过程中的Tafel曲线如图4(h)所示, $\text{Ni}_{0.1}\text{Zn}_{0.1}\text{Co}_{0.8}\text{Se}_2$ -S的Tafel曲线斜率最小, 而 $\text{Zn}_{0.2}\text{Co}_{0.8}\text{Se}_2$ 比 $\text{Ni}_{0.2}\text{Co}_{0.8}\text{Se}_2$ 的Tafel曲线斜率更小, 即Zn掺杂 CoSe_2 能更有效地促进 Li_2S 的分解, 降低了 Li_2S 的分解势垒。本工作详细地探讨了双掺杂电催化剂的协同作用, 并证实双掺杂比单掺杂具有更好的性能, 为锂硫催化关键材料的设计提供了一种新的策略。

2.3 异质结构调控

异质结构通常指由两种或两种以上的材料通过化学键耦合而形成的复合材料^[74,75]。异质结作为电催化剂可有效协同不同材料的优点, 显著提升催化活性^[76,77]。对硫正极而言, 因其催化活性高, 可有效促进多硫化物的吸附及转化; 而应用于负极一侧则可以促

进锂离子的均匀形核, 延缓锂枝晶的生长^[78].

对于正极侧的研究, 异质结构的设计通常更侧重于促进多硫化物的吸附转化. 以两种金属化合物构成的异质结受到广泛关注. 张亮课题组^[79]为改善单组分催化剂催化活性弱及催化稳定性差等问题, 原位制备了具有大量异质界面和丰富界面缺陷的 $\text{Co}_3\text{O}_4/\text{TiO}_2$ 异质结($\text{p-Co}_3\text{O}_4/\text{n-TiO}_2$ -HPs)作为硫宿主, 可有效实现硫还原过程中 S_8 、 Li_2S_4 、 Li_2S 三者之间的分步催化. DFT计算显示, 由于氧空位和多价金属位的存在, TiO_2 的(004)晶面对多硫化物具有更高的吸附能(图5(a)). 而在 Li_2S_8 还原为 Li_2S_4 过程中, Co_3O_4 与 TiO_2 相比, 前者具有更低的自由能垒. 因此, Co_3O_4 更有利于 Li_2S_8 的还原(图5(b)), 即 $\text{p-Co}_3\text{O}_4/\text{n-TiO}_2$ -HPs异质结对不同的多硫化物物种具有明显的定向催化作用, 高活性 Co_3O_4 片层保证了 S_8 的高度分散并精确诱导了固体 S_8 向液体 Li_2S_4 的

快速转化, 而嵌入的 TiO_2 则有助于强烈锚定 Li_2S_4 . 以 $\text{p-Co}_3\text{O}_4/\text{n-TiO}_2$ -HPs为硫宿主制备的硫正极($\text{S}@p\text{-Co}_3\text{O}_4/\text{n-TiO}_2$ -HPs)具有优异的倍率性能和循环稳定性. 其在10 C的大电流密度下, 初始容量为 610 mAh g^{-1} , 循环500圈后, 每圈衰减率仅为0.07%(图5(c)). 本研究揭示了异质结构在空间上的协同作用, 可以有效促进多硫化物的逐步转化, 为设计催化多硫化物氧化还原反应的界面结构提供了新思路.

异质结除了作为硫正极的电催化剂之外, 还可以有效调控锂金属的成核, 抑制锂负极的枝晶生长. 余彦研究组^[80]报道了一种原位嵌入 TiN-VN 异质结构的双功能柔性碳纳米纤维导电框架(TiN-VN@CNFs), 并将其同时用作硫正极(S/TiN-VN@CNFs)和锂负极(Li/TiN-VN@CNFs)的宿主. 高分辨透射电子显微镜(high-resolution transmission electron microscopy, HRTEM)图

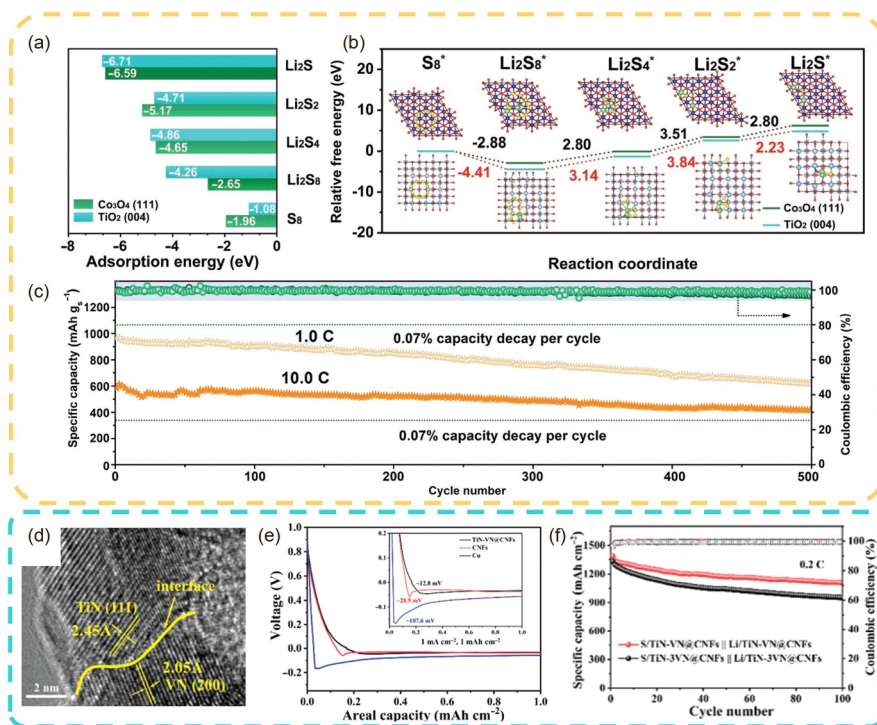


图 5 (网络版彩色) 锂硫电池中的异质结调控. 在 Co_3O_4 (111)晶面和 TiO_2 (004)晶面上, S_8 与 Li_2S_n ($n=1, 2, 4, 8$)的吸附能(a)和硫还原过程中的自由能(b). (c) $\text{S}@p\text{-Co}_3\text{O}_4/\text{n-TiO}_2$ -HPs正极在1.0和10.0 C下的长循环性能^[79], Copyright © 2021, John Wiley and Sons. (d) TiN-VN@CNFs 的高分辨透射电子显微镜(high-resolution transmission electron microscopy, HRTEM)图像. (e) 在电流密度为 1 mA cm^{-2} , 容量为 1 mAh cm^{-2} 的条件下, 3个不同电极在第一个循环沉积/剥离锂的电压-容量曲线. (f) 在 0.2 C 下 $\text{S/TiN-VN@CNFs}||\text{Li/TiN-VN@CNFs}$ 和 $\text{S/TiN-3VN@CNFs}||\text{Li/TiN-3VN@CNFs}$ 全电池的循环性能^[80], Copyright © 2020, John Wiley and Sons

Figure 5 (Color online) Heterostructure modulation in Li-S batteries. Adsorption energies of S_8 and different Li_2S_n ($n=1, 2, 4, 8$) species (a) and free energy profiles of the sulfur reduction process on the Co_3O_4 (111) and TiO_2 (004) surfaces (b). (c) Long-term cycling performance of $\text{S}@p\text{-Co}_3\text{O}_4/\text{n-TiO}_2$ -HPs cathode at 1.0 and 10.0 C^[79], Copyright © 2021, John Wiley and Sons. (d) HRTEM image of the TiN-VN@CNFs . (e) Voltage-capacity profiles of Li plating/stripping on three different current collectors for the first cycle at a current density of 1 mA cm^{-2} and a capacity of 1 mAh cm^{-2} . (f) Cycling performances of the $\text{S/TiN-VN@CNFs}||\text{Li/TiN-VN@CNFs}$ and the $\text{S/TiN-3VN@CNFs}||\text{Li/TiN-3VN@CNFs}$ full cells at 0.2 C ^[80], Copyright © 2020, John Wiley and Sons

像可以清楚地观察到TiN与VN的界面。这些丰富的异质界面作为活性位点可以有效促进界面电荷转移和表面吸附(图5(d))。作为硫宿主, TiN-VN@CNFs因其吸附性强且导电率高的特点, 并且TiN-VN具有优异的电催化活性, 因此可以有效抑制“穿梭效应”, 促进多硫化物的高效转化。此外, TiN-VN具有良好的亲锂性, 可降低锂金属的成核过电位, 有助于实现锂金属的均匀沉积并且抑制枝晶生长(图5(e))。因此, 这种正负极同时调控后组装的锂硫全电池(表示为S/TiN-VN@CNFs||Li/TiN-VN@CNFs)具有优异的电化学性能(图5(f))。该研究工作所展示的优异的电化学性能为实现高稳定、高能量密度以及长循环寿命的锂硫电池提供了有效的途径。

2.4 尺寸调控

减小颗粒尺寸可以有效增大电催化剂的比表面积, 从而暴露更多的活性位点, 大幅度提高电催化剂的催化效率^[81]。因而, 与纳米团簇相比, 具有更高比表面积和高原子利用率的单原子催化剂(single-atom catalyst, SAC)引起研究人员的广泛关注^[82,83]。尤其在锂硫电池体系中, 具有优异电催化效应的SAC对多硫化物吸附能力强并且可以有效改善氧化还原反应过程中动力学缓慢的问题^[84,85]。

其中, 以铁(Fe)^[86,87]、钴(Co)、钒(V)^[85]为金属中心的SAC修饰硫正极较为常见。Cai课题组^[88]基于DFT模拟了4种SAC材料TM-N₄-C(TM = Co, Fe, V, W)的电催化性能。作者首次对钨(W)为金属中心的SAC在锂硫电池中的应用进行了研究, 并与其他高性能SAC(以Co、Fe、V为金属中心)和石墨烯进行了比较。他们系统地研究了不同TM-N₄-C催化剂与Li₂S_n(n = 1, 2, 4, 6和8)五种多硫化物的结合能以及Li₂S₂转化为Li₂S的解离能。通过比较Li₂S_n在石墨烯和4种TM-N₄-C衬底上的吸附能可知, 所有SAC都表现出比石墨烯更强的结合能(图6(a))。由于Li₂S₂向Li₂S的转化反应动力学缓慢, 是硫还原反应的限速步骤。因此, Li₂S₂向Li₂S转化的能垒是影响锂硫电池容量的关键。基于此, 作者计算了不同SAC作用下该过程的反应活化能及模拟了不同SAC的原子结构。计算得知, 所有TM-N₄-C基底的活化能垒均显著低于石墨烯基底($E_a = 2.73$ eV)(图6(b)~(e))。进一步证明, SAC具有丰富的活性位点和优异的电催化活性, 可以有效降低反应活化能, 促进锂硫电池的氧化还原反应动力学。该工作详细探究了SAC对多硫化物

吸附和转化的催化作用, 证明了SAC在提高锂硫电池的电化学性能和缓解“穿梭效应”方面的潜力。

本课题组^[89]设计了一种Co-N_x修饰的碳纤维骨架(fibrous carbon skeleton implanted with single-atom dispersed Co-N_x species, Co-PCNF), 可同时实现抑制锂枝晶的生长和促进多硫化物的转化(图6(f)), 其可弯折的特性也为Co-PCNF用于柔性锂硫电池的构筑提供可能(图6(g))。具有亲锂性的Co-PCNF纤维骨架作为锂负极侧的中间层, 可以均匀电场和降低局部电流密度, 进而有效诱导锂离子的均匀沉积, 减缓锂枝晶的生长。Co-PCNF的扫描电子显微镜(scanning electron microscopy, SEM)图像和透射电子显微镜(transmission electron microscopy, TEM)图像清晰地显示, Co-PCNF为表面粗糙的纤维结构, 内部中空且多孔框架相互连接(图6(h), (i))。此外, HAADF-STEM图像及对应的X射线能谱仪(energy-dispersive X-ray spectroscopy, EDS)图像验证了Co元素、C元素和N元素在纤维结构中的均匀分布(图6(j))。而具有高催化活性的Co-PCNF作为硫宿主, 可以有效促进多硫化锂的转化, 缓解多硫化物的“穿梭效应”。DFT计算结果表明, Li₂S在Co-PCNF上的分解能垒远低于Li₂S在CNF上的分解能垒, 进一步证明单原子Co-N_x可有效催化Li₂S的转化分解反应(图6(k), (l))。此工作在纤维碳骨架中注入单原子, 实现了锂硫电池正负极的同时调控, 有助于推进SAC在锂硫电池中的实际应用。

2.5 相调控

与晶体材料相比, 非晶材料的表面原子呈非周期性排列, 使其具有更多的表面悬挂键与不饱和位点, 而这些表面缺陷被认为是催化活性位点^[90-92]。此外, 对锂硫电池而言, 非晶催化剂也可以更好地适应正极侧严重的体积膨胀问题^[93]。

王功名课题组和洪勋课题组^[94]合成了厚度约为10 nm的超薄非晶态CoO纳米片(amorphous CoO nanosheets, a-CoO NSs)作为正极硫宿主。该工作通过非晶化诱导CoO的表面电子调制, 优化Co原子的电子结构和配位环境, 显著提高了CoO对多硫化物的吸附能。a-CoO NSs和晶态CoO纳米片(crystalline CoO nanosheets, c-CoO NSs)的X射线吸收近边结构(X-ray absorption near-edge structure, XANES)光谱以及傅里叶变换扩展X射线吸收精细结构(Fourier-transformed extended X-ray absorption fine structure, FT-EXAFS)光谱

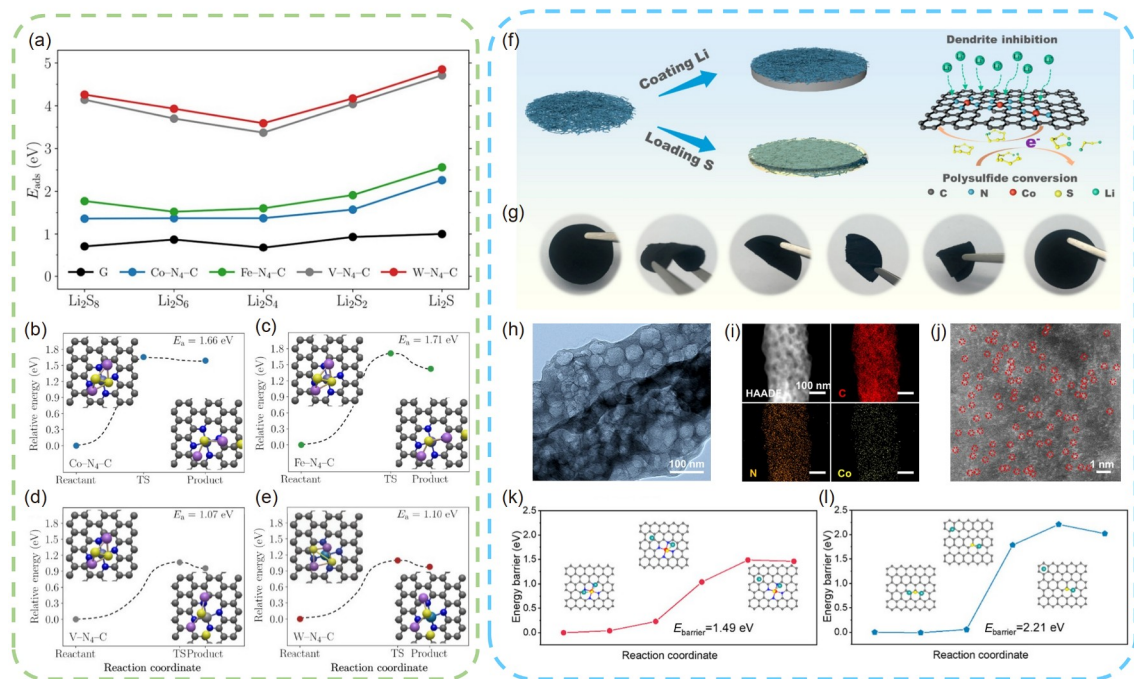


图6 (网络版彩色) 锂硫电池中的尺寸调控. (a) Li_2S_n 在纯石墨烯和SAC基底($\text{TM-N}_4\text{-C}$, $\text{TM} = \text{Co, Fe, V, W}$)上的吸附能. Co-N₄-C(b)、Fe-N₄-C(c)、V-N₄-C(d)、W-N₄-C(e)基底上 Li_2S_2 (反应物)转化为 Li_2S (生成物)的能垒及两者的原子结构^[88], Copyright © 2021, American Chemical Society. (f) 单原子Co-N_x修饰的双功能纤维骨架的设计原理示意图. (g) 纤维骨架在不同弯折状态下的数码图片. (h) Co-PCNF的SEM图像. (i) Co-PCNF的TEM图像. (j) Co-PCNF的HAADF-STEM图像及对应的EDS图像. 用DFT模拟计算了 Li_2S 在Co-PCNF(k)和CNF(l)上的分解能垒^[89], Copyright © 2021, American Chemical Society

Figure 6 (Color online) Size control of Li-S batteries. (a) Adsorption energies for Li_2S_n on pristine graphene and SAC substrates ($\text{TM-N}_4\text{-C}$, $\text{TM} = \text{Co, Fe, V, W}$). Dissociation energy graphs and atomic structures of Li_2S_2 (reactant) and Li_2S (product) for Co-N₄-C (b), Fe-N₄-C (c), V-N₄-C (d), and W-N₄-C (e) ^[88], Copyright © 2021, American Chemical Society. (f) Schematic diagram illustrating the structural configuration and design principle of the dual-functional fibrous skeleton harnessing single-atomic Co-N_x species. (g) Digital photos showing the flexibility and robustness of the fibrous skeleton. (h) SEM images of Co-PCNF. (i) TEM image of Co-PCNF. (j) HAADF-STEM image and the corresponding elemental maps of Co-PCNF. Decomposition energy barriers of Li_2S on Co-PCNF (k) and CNF (l) matrix simulated by the DFT method ^[89], Copyright © 2021, American Chemical Society

显示, a-CoO NSs和c-CoO NSs中Co的价态均为+2价(图7(a)), 且a-CoO NSs和c-CoO NSs皆具有与CoO相似的八面体结构(图7(b)). 此外, Co L-edge XANES对a-CoO中 Co^{2+} 阳离子的三维电子构型进行进一步研究显示, 与c-CoO NSs相比, a-CoO NSs的Co L-edge谱向低能区有明显的峰移(图7(c)). 为进一步探究 Co^{2+} 周围结构, 作者模拟了Co L-edge XANES的光谱, 结果表明, 非晶化处理可以部分改变Co原子周围配体场的对称性(图7(d)). 此外, DFT研究显示, c-CoO吸附 Li_2S_4 前后, 轨道分布变化不大(图7(e)), 而a-CoO吸附 Li_2S_4 后, 由于d轨道发生重排(图7(f)), 即非晶化诱导的d轨道再分布, 使更多的电子占据高能级, 从而与多硫化物具有较高的结合能. 该工作通过系统而深入地探讨a-CoO与c-CoO对吸附多硫化物前后的轨道分布变化, 提出了一种通过非晶态诱导表面电子态以显著提高电催化剂对多硫化物吸

附能力的策略, 这为合理调控电催化剂表面电子态以优化其在多硫体系中的吸附作用提供了新的指导.

除了设计非晶态材料直接作为高性能电催化剂以外, 将非晶材料负载到碳基基底上也是一种常见的形式. 郭林研究组^[95]设计了具有中空多孔结构的氮掺杂碳(NC)负载非晶 MoS_3 的纳米盒结构(NC/ MoS_3 NBs)作为正极硫宿主. 非晶态 MoS_3 的催化作用有效促进氧化还原反应动力学, 而NC使催化剂具有良好的电子导电性. 为印证NC/ MoS_3 NBs用于正极侧硫宿主的可行性, 将含硫的NC/ MoS_3 NBs(NC/ MoS_3 -S NBs)进行形貌表征. SEM图显示, NC/ MoS_3 -S NBs仍然保持原来均匀的立方形貌(图7(g)). HAADF-STEM图像显示, 立方壳层中颜色明显较亮部分为负载的硫(图7(h)), 这与图7(i)显示的线性扫描结果中硫的强信号相吻合. 此外, 非晶态NC/ MoS_3 NBs在提高硫负载量的同时, 也可以有效缓

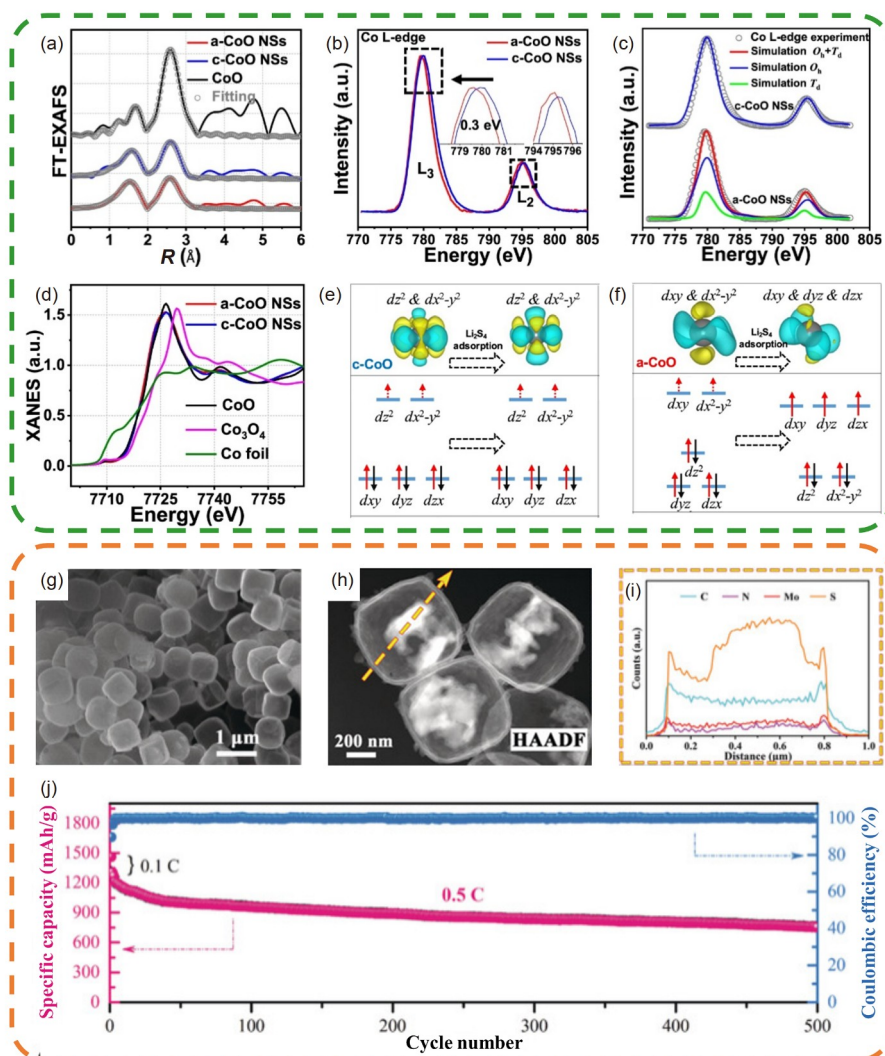


图 7 (网络版彩色) 锂硫电池中的相调控. (a) a-CoO NSs、c-CoO NSs、CoO、Co₃O₄ 和 Co 的 Co K-edge XANES 光谱. (b) a-CoO NSs、c-CoO NSs 和 CoO 的 FT-EXAFS 光谱. (c) a-CoO NSs 和 c-CoO NSs 的 Co L-edge XANES 光谱 (插图为 Co L₃ 和 L₂ 边的放大图像). (d) 模拟 a-CoO NSs 和 c-CoO NSs 的 Co L-edge XANES 光谱. (e) c-CoO 吸附 Li₂S₄ 前 (左) 后 (右) Co 的差分电荷密度示意图以及 Co 的 d 轨道示意图. (f) a-CoO 的吸附 Li₂S₄ 前 (左) 后 (右) Co 的差分电荷密度示意图以及 Co 的 d 轨道示意图^[94], Copyright © 2021, Nature Publishing Group. (g) NC/MoS₃-S NBs 的 SEM 图像. (h) NC/MoS₃-S NBs 的 HAADF-STEM 图像. (i) NC/MoS₃-S NBs 的元素线性分布图像. (j) NC/MoS₃-S NBs 正极在 0.5 C 下的长循环性能和库仑效率^[95], Copyright © 2020, John Wiley and Sons

Figure 7 (Color online) Phase modulation of Li-S batteries. (a) XANES spectra at the Co K-edge of the a-CoO NSs, c-CoO NSs, CoO, Co₃O₄ samples and Co foil. (b) FT-EXAFS spectra of the a-CoO NSs, c-CoO NSs, and CoO. (c) Co L-edge XANES spectra for the a-CoO NSs and c-CoO NSs (inset: The amplificatory image of Co L₃ and L₂ edge respectively). (d) Simulations of Co L-edge XANES spectra for the a-CoO NSs and c-CoO NSs. (e) The calculated deformation change density and the schematic of d orbitals for Co of c-CoO before (left) and after (right) Li₂S₄ adsorption. (f) The calculated deformation change density and the schematic of d orbitals for Co of a-CoO before (left) and after (right) Li₂S₄ adsorption^[94], Copyright © 2021, Nature Publishing Group. SEM (g), HAADF-STEM (h), and elemental linear distribution images (i) of NC/MoS₃-S NBs. (j) Long cycling performance and Coulombic efficiency of the NC/MoS₃-S NBs cathode at 0.5 C^[95], Copyright © 2020, John Wiley and Sons

解体积膨胀造成的电极开裂, 使 NC/MoS₃-S NBs 正极具有较高的比容量 (0.1 C 下的比容量为 1460 mAh g⁻¹)、优越的速率性能和长期循环性能 (在 0.5 C 下循环 500 次后, 容量保持为 752 mAh g⁻¹) (图 7(j)). 一方面, 该工作提供了一种可以同时实现物理约束和化学锚定的新型硫

宿主; 另一方面, 该合成策略可推广至其他非晶态金属硫化物纳米盒的制备.

3 总结与展望

如表 1 所示, 我们对非均相电催化剂的电池性能进

表 1 非均相电催化剂的电池性能对比

Table 1 Comparison of battery performances based on heterogeneous electrocatalysts

非均相电催化剂	非均相电催化剂调控策略	硫负载量 (mg cm ⁻²)	小倍率下的电池容量 (mAh g ⁻¹)	大倍率下的电池容量 (mAh g ⁻¹)	循环圈数(倍率)/衰减率(%)	高载量硫正极 (mg cm ⁻²)/ 电池容量 (mAh g ⁻¹)	参考文献
OV-T _n QDs	空位调控	2.2	1023.0 (0.1 C)	672.0 (2 C)	1000 (2 C)/0.012	4.8/736.0 (0.5 C)	[57]
Ni ₃ FeN/G	空位调控	1.2	1309.0 (0.1 C)	822.0 (3 C)		4.8/900.0 (0.5 C)	[58]
In ₂ S _{3-x} /rGO	空位调控		1202.0 (0.5 C)	705.0 (5.5 C)	450 (2 C)/0.100	4.0/539.0 (0.1 C)	[50]
WO@NC	空位调控	1.2–1.5	1078.0 (0.2 C)	688.0 (4 C)	1200 (2 C)/0.038	6.4/607.5 (0.2 C)	[60]
YP@SPB@P	掺杂调控	3.3	1322.0 (0.1 C)		50 (0.1 C)/0.19	8.1/027.0 (0.1 C)	[70]
Ni _{0.1} Zn _{0.1} Co _{0.8} Se ₂	掺杂调控	1.0	1219.4 (0.2 C)	681.7 (2 C)	400 (1 C)/0.065	4.6/598.0 (0.1 C)	[73]
Co/Co ₃ O ₄ -NHC	掺杂调控	1.2	957.1 (0.2 C)	447.9 (5 C)	500 (1 C)/0.032	4.0/371.9 (1 C)	[68]
N-PC@uCo/S	掺杂调控		1370.0 (0.1 C)	600.0 (5 C)	500 (1 C)/0.028	3.8/535.0 (0.5 C)	[67]
MSC/PCNiTe _{2-x}	掺杂调控	1.8	1318.0 (0.2 C)	764.0 (5 C)	1800 (4 C)/0.0139	7.3/890.4 (0.2 C)	[96]
p-Co ₃ O ₄ /n-TiO ₂ -HPs	异质结调控	1.0–1.3	1169.0 (0.2 C)	556.0 (10 C)	500 (10 C)/0.070	5.5/1000 (0.1 C)	[79]
TiN-VN@CNFs	异质结调控	1.7	1385.0 (0.2 C)	650.0 (5 C)	600 (2 C)/0.051	5.6/982.1 (0.1 C)	[80]
S/G-Nb ₂ O ₅	异质结调控	1.4–1.6	1188.0 (0.2 C)	768.0 (3 C)	1000 (2 C)/0.053	8.1/567.0 (0.2 C)	[78]
CoTe-MXene	异质结调控	1.0	1664.0 (0.1 C)	952.0 (2 C)	300 (2 C)	5.6/714.3 (0.02 C)	[97]
Co-N _x	尺寸调控	1.7	965.0 (0.1 C)	617.0 (1 C)	600 (0.2 C)/0.082	6.9/1036.2 (0.1 C)	[89]
3DFeSA-CN	尺寸调控	1.7	1363.5 (0.2 C)	783.0 (3 C)	2000 (1 C)/0.031	5.75/897.4 (0.2 C)	[87]
Co/SA-Zn@NC/CNTs	尺寸调控	1.0	1185.0 (0.2 C)	655.0 (5 C)	800 (1 C)/0.033	5.1/882.4 (0.2 C)	[84]
FeNSC	尺寸调控		1306.0 (0.05 C)	550.2 (4 C)	1000 (1 C)/0.047		[86]
Mn/Co-N-C	尺寸调控	1.0	1117.0 (0.2 C)	625.0 (6 C)	2000 (2 C)/0.036	14.1/798.6 (0.1 C)	[98]
Pt SAs/In ₂ S ₃ /Ti ₃ C ₂	尺寸调控	2.1–2.3	1115.1 (0.2 C)	719.6 (5 C)	1000 (0.5 C)/0.019	8.2/575.6 (0.2 C)	[99]
a-CoO NSs	相调控	2.0	1526.3 (0.2 C)	702.3 (5 C)	500 (1 C)/0.034		[94]
NC/MoS ₃ -S NBs	相调控	1.0	1267.2 (0.1 C)	596.7 (3 C)	500 (0.5 C)/0.076	5.5/643.6 (0.2 C)	[95]

行对比分析可知, 在锂硫电池中通过非均相电催化剂的引入, 可以有效促进反应动力学并提升其电化学性能。然而, 锂硫电池实用化的进程中仍有诸多问题亟待解决。首先, 缺乏高选择性且长效催化的电催化剂; 其次, 电催化剂中催化活性位点的催化机制仍不明确; 再者, 目前锂硫电池在贫锂贫液条件下性能不佳。针对以上问题, 为进一步提高锂硫电池中电催化剂的催化性能并明晰其催化机理, 我们提出以下值得探索的研究方向(图8)。

首先, 新兴的电催化剂仍需进一步设计与调控。以单原子催化剂为例, 将MXene、MoS₂等较为常见的二维材料作为单原子催化剂基底的电催化剂, 尚未被应用到锂硫电池体系中; 同时, 新兴的双单原子催化剂也鲜有应用。除本文评述的非均相催化剂以外, 多为溶解态小分子的均相催化剂具有活性位点暴露充分且构效关系清晰等优势, 有望在锂硫电池体系中获得广泛应用。

其次, 仍需加强电催化剂活性位点的精准合成与控制。与传统的纳米电催化剂的制备方法相比, 化学气



图 8 (网络版彩色)高比能、高稳定、高安全锂硫电池的未来发展方向

Figure 8 (Color online) Future development directions of high-energy, high-stability and high-safety Li-S batteries

相沉积以及原子层沉积等方法可以有效实现非均相催化材料的可控制备。然而, 针对锂硫电池中硫还原和硫析出反应过程, 如何设计具有针对性的催化活性位点仍有待研究。尤其是对于锂硫电池的多步转化反应过程, 若能实现选择性催化, 提高整体的反应速率, 将会进一步优化锂硫电池的电化学性能。

再次, 揭示非均相电催化剂催化机制的先进原位表征技术进一步发展。近年来, 原位TEM、原位拉曼、原位X射线衍射仪等表征技术在锂硫电池体系的广泛应用, 使得在电催化剂作用下多硫化物的转化过程逐渐明晰, 但电催化反应过程中电催化剂的结构、价态、配位信息仍不清晰。为进一步深入探究电催化剂

的催化机制以及非均相电催化剂在充放电过程中的原位演化过程, 仍需进一步发展如飞行时间二次离子质谱仪、原位同步辐射、原位固态核磁等先进表征技术。

最后, 亟待面向实用化的锂硫电池电催化剂的设计。高的锂金属和电解液用量以及正极侧高的导电碳含量严重阻碍了锂硫电池实现实际应用。为了获得高能量密度且可以稳定循环的锂硫电池, 一方面要设计高放电深度的锂金属负极, 以在贫锂、贫电解液条件下提升软包电池的循环稳定性, 另一方面还要降低正极侧导电碳的含量和正极孔隙率, 以减少电解液用量, 实现锂硫电池能量密度的提升。

参考文献

- 1 Ji X, Lee K T, Nazar L F. A highly ordered nanostructured carbon-sulphur cathode for lithium-sulphur batteries. *Nat Mater*, 2009, 8: 500–506
- 2 Liang X, Kwok C Y, Lodi-Marzano F, et al. Tuning transition metal oxide-sulfur interactions for long life lithium sulfur batteries: The “Goldilocks” principle. *Adv Energy Mater*, 2016, 6: 1501636
- 3 Liu Y, Meng X, Wang Z, et al. A Li₂S-based all-solid-state battery with high energy and superior safety. *Sci Adv*, 2022, 8: eabl8390
- 4 Peng H J, Huang J Q, Cheng X B, et al. Review on high-loading and high-energy lithium-sulfur batteries. *Adv Energy Mater*, 2017, 7: 1700260
- 5 Lim W G, Kim S, Jo C, et al. A comprehensive review of materials with catalytic effects in Li-S batteries: Enhanced redox kinetics. *Angew Chem Int Edit*, 2019, 58: 18746–18757
- 6 Song Y, Cai W, Kong L, et al. Rationalizing electrocatalysis of Li-S chemistry by mediator design: Progress and prospects. *Adv Energy Mater*, 2020, 10: 1901075
- 7 Tian Y, Li G, Zhang Y, et al. Low-bandgap Se-deficient antimony selenide as a multifunctional polysulfide barrier toward high-performance lithium-sulfur batteries. *Adv Mater*, 2020, 32: 1904876
- 8 Meng X, Sun Y, Yu M, et al. Hydrogen-bonding crosslinking MXene to highly robust and ultralight aerogels for strengthening lithium metal anode. *Small Sci*, 2021, 1: 2100021
- 9 Shi Z, Li M, Sun J, et al. Defect engineering for expediting Li-S chemistry: Strategies, mechanisms, and perspectives. *Adv Energy Mater*, 2021, 11: 2100332
- 10 Feng S, Fu Z, Chen X, et al. A review on theoretical models for lithium-sulfur battery cathodes. *InfoMat*, 2022, 4: e12304
- 11 Yu S, Cai W, Chen L, et al. Recent advances of metal phosphides for Li-S chemistry. *J Energy Chem*, 2021, 55: 533–548
- 12 Li Y, Gao T, Ni D, et al. Two birds with one stone: Interfacial engineering of multifunctional Janus separator for lithium-sulfur batteries. *Adv Mater*, 2022, 34: 2107638
- 13 Fang R, Zhao S, Sun Z, et al. More reliable lithium-sulfur batteries: Status, solutions and prospects. *Adv Mater*, 2017, 29: 1606832
- 14 Chung S, Chang C, Manthiram A. Progress on the critical parameters for lithium-sulfur batteries to be practically viable. *Adv Funct Mater*, 2018, 28: 1801188
- 15 Cai W, Song Y, Fang Y, et al. Defect engineering on carbon black for accelerated Li-S chemistry. *Nano Res*, 2020, 13: 3315–3320
- 16 He J, Manthiram A. A review on the status and challenges of electrocatalysts in lithium-sulfur batteries. *Energy Storage Mater*, 2019, 20: 55–70
- 17 Mao H, Liu L, Shi L, et al. High loading cotton cellulose-based aerogel self-standing electrode for Li-S batteries. *Sci Bull*, 2020, 65: 803–811
- 18 Wang H, Zhang W, Xu J, et al. Advances in polar materials for lithium-sulfur batteries. *Adv Funct Mater*, 2018, 28: 1707520
- 19 Kang H, Lee T, Kim H, et al. Thick free-standing electrode based on carbon-carbon nitride microspheres with large mesopores for high-energy-density lithium-sulfur batteries. *Carbon Energy*, 2021, 3: 410–423
- 20 Fan X, Sun W, Meng F, et al. Advanced chemical strategies for lithium-sulfur batteries: A review. *Green Energy Environ*, 2018, 3: 2–19
- 21 Ci H, Cai J, Ma H, et al. Defective VSe₂-graphene heterostructures enabling *in situ* electrocatalyst evolution for lithium-sulfur batteries. *ACS Nano*, 2020, 14: 11929–11938
- 22 Jin J, Cai W, Cai J, et al. MOF-derived hierarchical CoP nanoflakes anchored on vertically erected graphene scaffolds as self-supported and flexible hosts for lithium-sulfur batteries. *J Mater Chem A*, 2020, 8: 3027–3034

- 23 Deng C, Wang Z, Feng L, et al. Electrocatalysis of sulfur and polysulfides in Li-S batteries. *J Mater Chem A*, 2020, 8: 19704–19728
- 24 Lin Y, Huang S, Zhong L, et al. Organic liquid electrolytes in Li-S batteries: Actualities and perspectives. *Energy Storage Mater*, 2021, 34: 128–147
- 25 Shi Z, Ding Y, Zhang Q, et al. Electrocatalyst modulation toward bidirectional sulfur redox in Li-S batteries: From strategic probing to mechanistic understanding. *Adv Energy Mater*, 2022, 12: 2201056
- 26 Ng S F, Lau M Y L, Ong W J. Lithium-sulfur battery cathode design: Tailoring metal-based nanostructures for robust polysulfide adsorption and catalytic conversion. *Adv Mater*, 2021, 33: 2008654
- 27 Li J, Niu Z, Guo C, et al. Catalyzing the polysulfide conversion for promoting lithium sulfur battery performances: A review. *J Energy Chem*, 2021, 54: 434–451
- 28 Zhang M, Chen W, Xue L, et al. Adsorption-catalysis design in the lithium-sulfur battery. *Adv Energy Mater*, 2020, 10: 1903008
- 29 Geng P, Du M, Guo X, et al. Bimetallic metal-organic framework with high-adsorption capacity toward lithium polysulfides for lithium-sulfur batteries. *Energy Environ Mater*, 2021, 5: 599–607
- 30 Zhou L, Danilov D L, Eichel R, et al. Host materials anchoring polysulfides in Li-S batteries reviewed. *Adv Energy Mater*, 2021, 11: 2001304
- 31 Shi Z, Sun Z, Cai J, et al. Manipulating electrocatalytic Li_2S redox via selective dual-defect engineering for Li-S batteries. *Adv Mater*, 2021, 33: 2103050
- 32 Jiang J, Fan Q, Zheng Z, et al. The dual functions of defect-rich carbon nanotubes as both conductive matrix and efficient mediator for Li-S batteries. *Small*, 2021, 17: 2103535
- 33 Zhao C X, Chen W J, Zhao M, et al. Redox mediator assists electron transfer in lithium-sulfur batteries with sulfurized polyacrylonitrile cathodes. *EcoMat*, 2020, 3: e12066
- 34 Fang R, Chen K, Yin L, et al. The regulating role of carbon nanotubes and graphene in lithium-ion and lithium-sulfur batteries. *Adv Mater*, 2019, 31: 1800863
- 35 Song Y, Sun Z, Fan Z, et al. Rational design of porous nitrogen-doped Ti_3C_2 MXene as a multifunctional electrocatalyst for Li-S chemistry. *Nano Energy*, 2020, 70: 104555
- 36 Sun R, Bai Y, Bai Z, et al. Phosphorus vacancies as effective polysulfide promoter for high-energy-density lithium-sulfur batteries. *Adv Energy Mater*, 2022, 12: 2102739
- 37 Hou W, Feng P, Guo X, et al. Catalytic mechanism of oxygen vacancies in perovskite oxides for lithium-sulfur batteries. *Adv Mater*, 2022, 34: 2202222
- 38 Luo D, Zhang Z, Li G, et al. Revealing the rapid electrocatalytic behavior of ultrafine amorphous defective $\text{Nb}_2\text{O}_{5-x}$ nanocluster toward superior Li-S performance. *ACS Nano*, 2020, 14: 4849–4860
- 39 Ci H, Wang M, Sun Z, et al. Direct insight into sulfiphilicity-lithiophilicity design of bifunctional heteroatom-doped graphene mediator toward durable Li-S batteries. *J Energy Chem*, 2022, 66: 474–482
- 40 Gao X, Yang X, Li M, et al. Cobalt-doped SnS_2 with dual active centers of synergistic absorption-catalysis effect for high-S loading Li-S batteries. *Adv Funct Mater*, 2019, 29: 1806724
- 41 Sun T, Zhao X, Li B, et al. NiMoO_4 nanosheets anchored on N-S doped carbon clothes with hierarchical structure as a bidirectional catalyst toward accelerating polysulfides conversion for Li-S battery. *Adv Funct Mater*, 2021, 31: 2101285
- 42 Wang S, Liu X, Duan H, et al. $\text{Fe}_3\text{C}/\text{Fe}$ nanoparticles embedded in N-doped porous carbon nanosheets and graphene: A thin functional interlayer for PP separator to boost performance of Li-S batteries. *Chem Eng J*, 2021, 415: 129001
- 43 Ren Y, Zhai Q, Wang B, et al. Synergistic adsorption-electrocatalysis of 2D/2D heterostructure toward high performance Li-S batteries. *Chem Eng J*, 2022, 439: 135535
- 44 Xu J, Xu L, Zhang Z, et al. Heterostructure ZnSe-CoSe_2 embedded with yolk-shell conductive dodecahedral as two-in-one hosts for cathode and anode protection of lithium-sulfur full batteries. *Energy Storage Mater*, 2022, 47: 223–234
- 45 Li Y, Wu J, Zhang B, et al. Fast conversion and controlled deposition of lithium (poly)sulfides in lithium-sulfur batteries using high-loading cobalt single atoms. *Energy Storage Mater*, 2020, 30: 250–259
- 46 Xie S, Chen X, Wang C, et al. Role of the metal atom in a carbon-based single-atom electrocatalyst for Li-S redox reactions. *Small*, 2022, 18: 2200395
- 47 Li Y, Lin S, Wang D, et al. Single atom array mimic on ultrathin MOF nanosheets boosts the safety and life of lithium-sulfur batteries. *Adv Mater*, 2020, 32: 1906722
- 48 Zhang X, Li G, Zhang Y, et al. Amorphizing metal-organic framework towards multifunctional polysulfide barrier for high-performance lithium-sulfur batteries. *Nano Energy*, 2021, 86: 106094
- 49 Wang M, Sun Z, Ci H, et al. Identifying the evolution of selenium-vacancy-modulated MoSe_2 precatalyst in lithium-sulfur chemistry. *Angew Chem Int Edit*, 2021, 60: 24558–24565

- 50 He D, Liu J, Zhang B, et al. Enhancing adsorption and catalytic activity of marigold-like In_2S_3 in lithium-sulfur batteries by vacancy modification. *Chem Eng J*, 2022, 427: 131711
- 51 Zhang X, Liu X, Zhang W, et al. Tunable vacancy defect chemistry on free-standing carbon cathode for lithium-sulfur batteries. *Green Energy Environ*, 2022, doi: 10.1016/j.gee.2022.03.006
- 52 Li C, Zhao Y, Zhang Y, et al. A new defect-rich and ultrathin ZnCo layered double hydroxide/carbon nanotubes architecture to facilitate catalytic conversion of polysulfides for high-performance Li-S batteries. *Chem Eng J*, 2021, 417: 129248
- 53 Xu L, Zhao H, Sun M, et al. Oxygen vacancies on layered niobic acid that weaken the catalytic conversion of polysulfides in lithium-sulfur batteries. *Angew Chem Int Edit*, 2019, 58: 11491–11496
- 54 Li Z, Zhang Q, Hencz L, et al. Multifunctional cation-vacancy-rich ZnCo_2O_4 polysulfide-blocking layer for ultrahigh-loading Li-S battery. *Nano Energy*, 2021, 89: 106331
- 55 Luo D, Li G, Deng Y, et al. Synergistic engineering of defects and architecture in binary metal chalcogenide toward fast and reliable lithium-sulfur batteries. *Adv Energy Mater*, 2019, 9: 1900228
- 56 Song Y, Gao H, Wang M, et al. Deciphering the defect micro-environment of graphene for highly efficient Li-S redox reactions. *EcoMat*, 2022, 4: e12182
- 57 Zhang H, Yang L, Zhang P, et al. MXene-derived $\text{Ti}_n\text{O}_{2n-1}$ quantum dots distributed on porous carbon nanosheets for stable and long-life Li-S batteries: Enhanced polysulfide mediation via defect engineering. *Adv Mater*, 2021, 33: 2008447
- 58 Zhao M, Peng H, Zhang Z, et al. Activating inert metallic compounds for high-rate lithium-sulfur batteries through *in situ* etching of extrinsic metal. *Angew Chem Int Edit*, 2019, 58: 3779–3783
- 59 Jin H G, Wang M, Wen J X, et al. Oxygen vacancy-rich mixed-valence cerium MOF: An efficient separator coating to high-performance lithium-sulfur batteries. *ACS Appl Mater Interfaces*, 2021, 13: 3899–3910
- 60 Wang S, Wang Y, Song Y, et al. Immobilizing polysulfide via multiple active sites in $\text{W}_{18}\text{O}_{49}$ for Li-S batteries by oxygen vacancy engineering. *Energy Storage Mater*, 2021, 43: 422–429
- 61 Wang J, Han W. A review of heteroatom doped materials for advanced lithium-sulfur batteries. *Adv Funct Mater*, 2022, 32: 2107166
- 62 Ren X, Liu Z, Zhang M, et al. Review of cathode in advanced Li-S batteries: The effect of doping atoms at micro levels. *ChemElectroChem*, 2021, 8: 3457–3471
- 63 Saroha R, Oh J H, Lee J S, et al. Hierarchically porous nanofibers comprising multiple core-shell Co_3O_4 @graphitic carbon nanoparticles grafted within N-doped CNTs as functional interlayers for excellent Li-S batteries. *Chem Eng J*, 2021, 426: 130805
- 64 Su H, Lu L, Yang M, et al. Decorating CoSe_2 on N-doped carbon nanotubes as catalysts and efficient polysulfides traps for Li-S batteries. *Chem Eng J*, 2022, 429: 132167
- 65 Wang P, Sun F, Xiong S, et al. WSe_2 flakelets on N-doped graphene for accelerating polysulfide redox and regulating Li plating. *Angew Chem Int Edit*, 2022, 61: e202116048
- 66 Xiao Q, Yang J, Wang X, et al. Carbon-based flexible self-supporting cathode for lithium-sulfur batteries: Progress and perspective. *Carbon Energy*, 2021, 3: 271–302
- 67 Wang R, Yang J, Chen X, et al. Highly dispersed cobalt clusters in nitrogen-doped porous carbon enable multiple effects for high-performance Li-S battery. *Adv Energy Mater*, 2020, 10: 1903550
- 68 Jeon Y, Lee J, Jo H, et al. $\text{Co}/\text{Co}_3\text{O}_4$ -embedded N-doped hollow carbon composite derived from a bimetallic MOF/ ZnO core-shell template as a sulfur host for Li-S batteries. *Chem Eng J*, 2021, 407: 126967
- 69 Feng T, Zhao T, Zhu S, et al. Anion-doped cobalt selenide with porous architecture for high-rate and flexible lithium-sulfur batteries. *Small Methods*, 2021, 5: 2100649
- 70 Jo S C, Hong J W, Choi I H, et al. Multimodal capturing of polysulfides by phosphorus-doped carbon composites for flexible high-energy-density lithium-sulfur batteries. *Small*, 2022, 18: 2200326
- 71 Marangon V, Di Lecce D, Brett D J L, et al. Characteristics of a gold-doped electrode for application in high-performance lithium-sulfur battery. *J Energy Chem*, 2022, 64: 116–128
- 72 Hu F, Peng H, Zhang T, et al. A lightweight nitrogen/oxygen dual-doping carbon nanofiber interlayer with meso-/micropores for high-performance lithium-sulfur batteries. *J Energy Chem*, 2021, 58: 115–123
- 73 Chen L, Xu Y, Cao G, et al. Bifunctional catalytic effect of CoSe_2 for lithium-sulfur batteries: Single doping versus dual doping. *Adv Funct Mater*, 2022, 32: 2107838
- 74 Huang S, Wang Z, Von Lim Y, et al. Recent advances in heterostructure engineering for lithium-sulfur batteries. *Adv Energy Mater*, 2021, 11: 2003689
- 75 Li Y, Zhang J, Chen Q, et al. Emerging of heterostructure materials in energy storage: A review. *Adv Mater*, 2021, 33: 2100855
- 76 Zhang H, Liu Q, Ruan S, et al. *In-situ* construction of $\text{g-C}_3\text{N}_4$ /carbon heterostructure on graphene nanosheet: An efficient polysulfide barrier for

- advanced lithium-sulfur batteries. *Appl Surf Sci*, 2022, 578: 152022
- 77 Nguyen T T, Balamurugan J, Go H W, et al. Dual-functional $\text{Co}_{5.47}\text{N}/\text{Fe}_3\text{N}$ heterostructure interconnected 3D N-doped carbon nanotube-graphene hybrids for accelerating polysulfide conversion in Li-S batteries. *Chem Eng J*, 2022, 427: 131774
- 78 Wang M, Song Y, Wei N, et al. Universal interface and defect engineering dual-strategy for graphene-oxide heterostructures toward promoted Li-S chemistry. *Chem Eng J*, 2021, 418: 129407
- 79 Li H, Chen C, Yan Y, et al. Utilizing the built-in electric field of p-n junctions to spatially propel the stepwise polysulfide conversion in lithium-sulfur batteries. *Adv Mater*, 2021, 33: 2105067
- 80 Yao Y, Wang H, Yang H, et al. A dual-functional conductive framework embedded with TiN-VN heterostructures for highly efficient polysulfide and lithium regulation toward stable Li-S full batteries. *Adv Mater*, 2020, 32: 1905658
- 81 Liang Z, Shen J, Xu X, et al. Advances in the development of single-atom catalysts for high-energy-density lithium-sulfur batteries. *Adv Mater*, 2022, 34: 2200102
- 82 Zhang D, Wang S, Hu R, et al. Catalytic conversion of polysulfides on single atom zinc implanted MXene toward high-rate lithium-sulfur batteries. *Adv Funct Mater*, 2020, 30: 2002471
- 83 Han Z, Zhao S, Xiao J, et al. Engineering d-p orbital hybridization in single-atom metal-embedded three-dimensional electrodes for Li-S batteries. *Adv Mater*, 2021, 33: 2105947
- 84 Wang R, Wu R, Yan X, et al. Implanting single Zn atoms coupled with metallic Co nanoparticles into porous carbon nanosheets grafted with carbon nanotubes for high-performance lithium-sulfur batteries. *Adv Funct Mater*, 2022, 32: 2200424
- 85 Fang L, Feng Z, Cheng L, et al. Design principles of single atoms on carbons for lithium-sulfur batteries. *Small Methods*, 2020, 4: 2000315
- 86 Zhao H, Tian B, Su C, et al. Single-atom iron and doped sulfur improve the catalysis of polysulfide conversion for obtaining high-performance lithium-sulfur batteries. *ACS Appl Mater Interfaces*, 2021, 13: 7171–7177
- 87 Ding Y, Cheng Q, Wu J, et al. Enhanced dual-directional sulfur redox via a biotemplated single-atomic Fe-N₂ mediator promises durable Li-S batteries. *Adv Mater*, 2022, 34: 2202256
- 88 Andritsos E I, Lekakou C, Cai Q. Single-atom catalysts as promising cathode materials for lithium-sulfur batteries. *J Phys Chem C*, 2021, 125: 18108–18118
- 89 Huang T, Sun Y, Wu J, et al. A dual-functional fibrous skeleton implanted with single-atomic Co-N_x dispersions for longevous Li-S full batteries. *ACS Nano*, 2021, 15: 14105–14115
- 90 Li L, Shao Q, Huang X. Amorphous oxide nanostructures for advanced electrocatalysis. *Chem Eur J*, 2020, 26: 3943–3960
- 91 Anantharaj S, Noda S. Amorphous catalysts and electrochemical water splitting: An untold story of harmony. *Small*, 2020, 16: 1905779
- 92 Bergmann A, Martinez-Moreno E, Teschner D, et al. Reversible amorphization and the catalytically active state of crystalline Co_3O_4 during oxygen evolution. *Nat Commun*, 2015, 6: 8625
- 93 Xia G, Zhang L, Ye J, et al. Amorphous transformation of FeP enabling enhanced sulfur catalysis and anchoring in high-performance Li-S batteries. *Chem Eng J*, 2022, 431: 133705
- 94 Li R, Rao D, Zhou J, et al. Amorphization-induced surface electronic states modulation of cobaltous oxide nanosheets for lithium-sulfur batteries. *Nat Commun*, 2021, 12: 3102
- 95 Yu J, Xiao J, Li A, et al. Enhanced multiple anchoring and catalytic conversion of polysulfides by amorphous MoS_3 nanoboxes for high-performance Li-S batteries. *Angew Chem Int Edit*, 2020, 59: 13071–13078
- 96 Yao W, Tian C, Yang C, et al. P-doped NiTe_2 with Te-vacancies in lithium-sulfur batteries prevents shuttling and promotes polysulfide conversion. *Adv Mater*, 2022, 34: e2106370
- 97 Ye Z, Jiang Y, Li L, et al. Enhanced catalytic conversion of polysulfide using 1D CoTe and 2D MXene for heat-resistant and lean-electrolyte Li-S batteries. *Chem Eng J*, 2022, 430: 132734
- 98 Qiao S, Wang Q, Lei D, et al. Oxygen vacancy enabled fabrication of dual-atom Mn/Co catalysts for high-performance lithium-sulfur batteries. *J Mater Chem A*, 2022, 10: 11702–11711
- 99 Zhou C, Li M, Hu N, et al. Single-atom-regulated heterostructure of binary nanosheets to enable dendrite-free and kinetics-enhanced Li-S batteries. *Adv Funct Mater*, 2022, 33: 2204635

Summary for “锂硫电池非均相电催化剂”

Heterogeneous electrocatalysts for lithium-sulfur batteries

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With the rapid development of electronic transportation and portable electronics, commercial lithium-ion batteries would be unsatisfactory to meet the needs of future markets owing to the relatively low energy density. Seeking a reliable substitution system is becoming imperative. Lithium-sulfur (Li-S) battery has been recognized as a cutting-edge energy storage system because of its high theoretical capacity (1672 mAh g^{-1}) and cost-effectiveness of sulfur resources. Nevertheless, the severe shuttle effect and sluggish reaction kinetics significantly restrict its further commercialization. In the developing history of Li-S batteries, physical confinement and chemical adsorption have often been employed to achieve the anchoring of polysulfides. However, these strategies are still difficult to fully solve the problems of slow reaction kinetics and poor cycle performance. Recent years have witnessed the introduction of heterogeneous electrocatalysts into the Li-S system, which readily meets with great success in suppressing the shuttle effect and elevating the reaction kinetics.

In this review, we introduce the basic reaction of Li-S battery and point out the main challenges in its current development stage, followed by summarizing the effects of electrocatalysts. Firstly, the polar electrocatalyst is able to effectively adsorb polysulfide and inhibit its diffusion in the electrolyte. Secondly, the electrocatalyst can effectively promote the conversion of polysulfide and improve the utilization rate of sulfur species. Thirdly, the uniform distribution of active sites in the electrocatalyst also contributes to the uniform deposition of short-chain polysulfides. In addition, the concepts and comparisons of homogeneous and heterogeneous electrocatalysts are presented, manifesting the essential significance of heterogeneous electrocatalysts in Li-S system. Homogeneous electrocatalysts usually refer to small soluble organic molecules with the same phase state as soluble polysulfides in the reaction system. Homogeneous electrocatalysts have more specific catalytic sites, whilst heterogeneous electrocatalysts have more diverse morphologies and higher possibility of multi-site synergistic catalysis. Then the common regulation strategies for heterogeneous electrocatalysts are classified, including vacancy regulation, doping regulation, heterostructure modulation, size control, and phase modulation. In further contexts, the underlying mechanisms of different regulation strategies for promoting polysulfide conversion are discussed.

The introduction of heterogeneous electrocatalysts can effectively promote the reaction kinetics and improve their electrochemical performance. However, there are still many obstacles waiting to be solved in the practicability of Li-S batteries. For instance, there is a lack of highly selective and stable electrocatalysts. The identification of real catalytic sites in the electrocatalysts still remains unclear. Moreover, Li-S batteries have poor performance under lean electrolyte and low lithium consumption. In view of these issues, future development strategy is put forward from four aspects, encompassing material exploration, precise synthesis, characterization methods and practical application, aiming to bridge the gap between the reality and ideal in working Li-S batteries.

lithium-sulfur battery, heterogeneous electrocatalyst, shuttle effect, reaction kinetics

doi: [10.1360/TB-2022-0680](https://doi.org/10.1360/TB-2022-0680)